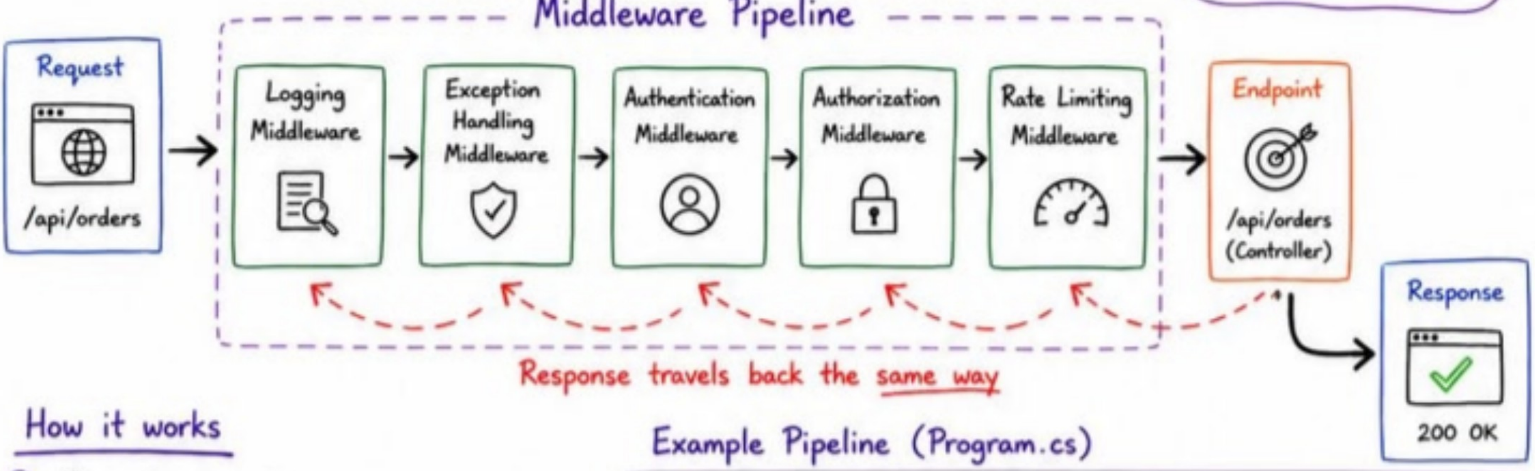
Middleware Pipeline in ASP.NET Core

Order matters.
First in, First to handle the request.
Last out,
First to respond.

Think of Middleware as a Highway for Requests



How it works

1. Request comes in
2. Each middleware can:
   - Do something (before next)
   - Call next() to pass to the next middleware
   - Do something again (after next)
3. Request reaches the endpoint
4. Response starts coming back
5. Each middleware can:
   - Do something (after next)
   - Modify the response
6. Response is sent to the client

Example Pipeline (Program.cs)

```
var builder = WebApplication.CreateBuilder(args);
var app = builder.Build();

app.UseHttpsRedirection();          // 1. Redirect to HTTPS
app.UseLogging();                   // 2. Log every request
app.UseExceptionHandler("/error");  // 3. Handle unhandled exceptions
app.UseAuthentication();            // 4. Validate user
app.UseAuthorization();             // 5. Check user permissions
app.UseRateLimiter();               // 6. Prevent abuse

app.MapControllers();               // 7. Endpoint / Controllers

app.Run();
```

Why Order Matters?

Wrong Order (Issues)

Auth → Exception Handler → Logging → ...

- Exceptions thrown before the handler won't be caught
- Logs may miss important request details
- Users might be challenged even for static files
- Rate limiting might run too late

Recommended Order

Logging → Exception Handler → HTTPS Redirection → Static Files → Routing → Auth → Authorization → Rate Limiting → Endpoint

General Rule:

- Cross-cutting concerns first (logging, errors)
- Security in the middle (auth, authorization)
- Business & endpoints at the end

Key Takeaways

- Middleware is the backbone of ASP.NET Core request handling.
- It's a chain. Each piece decides what happens next.
- Order is everything. A small change can break your flow.
- Keep it focused, small and single-purpose.
- Use built-in middleware whenever possible. Don't reinvent the wheel.

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